Low-Carbon and Energy-Saving Design of an Office Building in Chongqing

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Undergraduate Thesis, English Summary

This report is an English summary of the author's undergraduate thesis at Chongqing University (2025). The full thesis is in Simplified Chinese and is included in the project repository.

Abstract

Office buildings in Chongqing face long, humid summers and damp, cold winters, and many existing buildings have had no energy retrofit. This study develops an integrated low-carbon design for a six-storey office building in Chongqing using Ecotect for solar analysis and DeST-C for hourly simulation of indoor temperature, loads, and energy use. In the baseline, built to the national minimum standard, the open-plan office is within the 18–28 °C comfort band for only 41.8% of the year without air conditioning, and the energy use intensity (EUI) is 105.76 kWh/m² per year. A month-by-month ventilation schedule, interior blinds, and a better envelope reduce the EUI to 88.10 kWh/m² (−16.7%). Higher-efficiency heat pumps, lighting, and office equipment alone reduce it to 65.40 kWh/m² (−38.2%). Together, the measures lower the EUI to 55.28 kWh/m² (−47.7%), raise free-running comfortable hours by 65.3%, and save 269,338 kWh and 160.5 t CO_2 a year. A 267.8 kW rooftop photovoltaic system generating about 150,000 kWh a year covers all lighting electricity and pays back in about 9.5 years. The study also classifies building loads as rigid, reducible, or transferable to frame demand flexibility.

1. Introduction

Building operation accounts for about one-fifth of China's energy use, and public buildings are the largest share of it and still growing. Chongqing lies in the hot-summer, cold-winter climate zone, where offices need both cooling and heating, and the city keeps adding office space: in 2024 it completed 2.47 million m² of office buildings and had 47.13 million m² under construction, and many existing office buildings have had neither an energy retrofit nor energy management.

The thesis asks how far a typical Chongqing office building can cut its energy use and operational carbon emissions by combining renewable generation, passive design, efficient systems, and flexible operation, and which measures matter most. It answers the question with a case study of one building, simulated hour by hour for a full year in a baseline and in several improved designs.

2. Building and Climate

The case building is a six-storey office building with a floor height of 3.7 m, a trapezoidal footprint of 1,153 m², and a total floor area of 6,918 m², of which 5,236 m² is air-conditioned. Each floor has the same layout: an open-plan office of about 681 m², small and large meeting rooms, private offices, a restroom, and two stair and elevator cores.

Chongqing has a humid subtropical monsoon climate. The annual mean temperature is 17.7 °C, the mean relative humidity is 79.9%, the mean wind speed is only 1.1 m/s with calms 45.5% of the time, and there are about 1,156 hours of sunshine a year. Annual horizontal solar irradiation is below 1,050 kWh/m², the lowest class in China. DeST-C gives 1,102.8 heating degree-days and 183.4 cooling degree-days, but the cooling season is intense: on the cooling design day only four early-morning hours fall below 28 °C.

3. Methods

The building geometry was modeled in Revit and analyzed in two tools. Ecotect (with its Weather Tool) was used for the optimal orientation and the solar resource on the roof. DeST-C, a building simulation program developed at Tsinghua University and widely used in China, was used for hourly free-running indoor temperature, heating and cooling loads, and electricity use over a typical year.

The baseline follows the minimum requirements of the national standard for public buildings (GB 50189-2015) for the hot-summer, cold-winter zone (Table 1). Occupant heat gains follow ASHRAE Standard 55 metabolic rates by activity, and occupancy, lighting, and equipment schedules follow the standard's office profiles. Air conditioning runs from 7:00 to 18:00 on weekdays with a fan-coil and fresh-air system, a minimum of 30 m³/h of fresh air per person, and set points of 18 °C and 28 °C. Heating and cooling are supplied by air-source heat pumps with a coefficient of performance (COP) of 2.0.

Table 1. Baseline Building Parameters

Item	Baseline value
External wall	U = 0.796 W/(m²·K)
Roof	U = 0.493 W/(m²·K)
Windows (window-to-wall ratio 0.31)	U = 2.6 W/(m²·K), SHGC 0.75
Lighting power density (offices)	9 W/m²
Equipment power density (offices)	15 W/m²
Natural ventilation	1 air change per hour, all year
Heat pump COP	2.0

Energy use intensity. The thesis reports EUI as the sum of three end-use intensities: lighting electricity per lit floor area (6,015 m²), and equipment and heating and cooling electricity per air-conditioned area (5,236 m²). Heating and cooling electricity is the annual heating and cooling load divided by the heat pump COP. The same convention is used throughout, so the relative savings are comparable across designs. Dividing total electricity by the gross floor area of 6,918 m² instead gives 82.4 kWh/m² for the baseline and 43.5 kWh/m² for the integrated design, a 47.2% reduction.

4. Rooftop Photovoltaics

Ecotect places the best orientation for this climate at 7.5° east of south. A 30° panel tilt gave only about 5% more irradiation than horizontal mounting, and diffuse light dominates in Chongqing's cloudy climate, so the design uses horizontally mounted panels. The system has 412 bifacial double-glass modules of 650 W each (267.8 kW, 23.3% efficiency) on 1,151.67 m² of roof.

With the simulated roof irradiation of 700 kWh/m² per year and a performance factor of 0.8, the system generates 150,270 kWh a year, more than the baseline lighting demand of 125,837 kWh. A rated-capacity estimate based on sunshine hours gives 247,747 kWh, but the irradiation-based figure is the more realistic one for Chongqing and is used in the rest of the study. At a commercial tariff of 0.82 CNY/kWh, the system saves about 123,200 CNY a year. With an installed cost of about 902,000 CNY and

annual maintenance of 26,780 CNY, the simple payback period is 9.5 years (5.1 years with the rated-capacity estimate), and the system keeps 87.4% of its rated power after 30 years.

5. Baseline Performance

Without air conditioning, the open-plan office on a typical floor is within 18–28 °C for 3,662 hours (41.8%) of the year, mostly in spring and autumn. It is above 30 °C for 45.0% of the year and below 18 °C for 5.6%. The annual cooling load (78.01 kWh/m²) is more than four times the heating load (17.67 kWh/m²). The cooling load peaks at 1,030.94 kW on a July afternoon, with a single daily peak between 14:00 and 17:00, while the heating load peaks at 683.74 kW on a February morning and shows morning and evening peaks.

The baseline uses 570,034 kWh of electricity a year: 125,837 kWh for lighting, 193,714 kWh for equipment, and 250,484 kWh for heating and cooling. Its EUI is 105.76 kWh/m², of which heating and cooling account for 45.2%.

6. Passive Measures

6.1 Seasonal ventilation schedule

The baseline ventilates at one air change per hour all year, which is too little in spring and autumn and too much in winter. The thesis simulated seven constant rates from 0.5 (windows closed) to 10 air changes per hour and compared the heating, comfortable, and cooling hours of the open-plan office month by month. The best rate for each month forms the schedule in Table 2. To avoid drafts and blown papers, rates during working hours are capped at 5 air changes per hour; 10 air changes per hour are used only at night, when the building is empty, to purge heat.

Table 2. Seasonal Ventilation Schedule (Air Changes per hour)

Months	Working hours	Other hours
January–March, November–December	0.5	0.5
April, October	2.0	2.0
May, September	5.0	10.0
June–August	0.5 (air conditioning on)	10.0

The schedule cuts the hours that need heating from 1,572 to 307 (−80.5%) and the hours that need cooling from 3,793 to 1,317 (−65.3%), and more than doubles the comfortable hours, from 3,395 to 7,136 (Figure 1). Spring and autumn without air conditioning extend from March–April and September–

October to March–May and September–November. On its own, the schedule lowers the EUI to 95.31 kWh/m² (−9.9%).

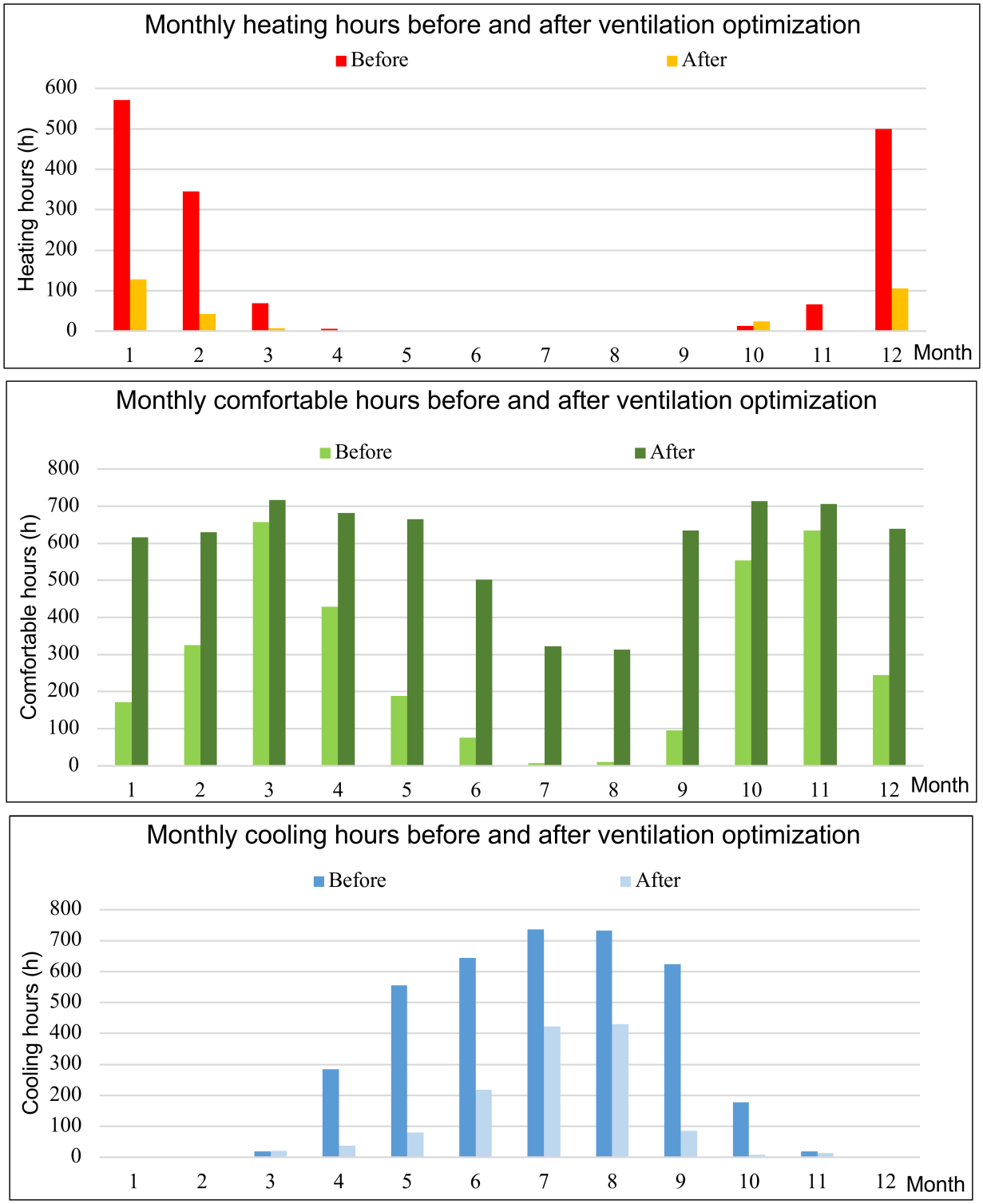


Figure 1. Monthly Heating (top), Comfortable (middle), and Cooling (bottom) Hours of the Open-plan Office on a Typical Floor without Air Conditioning, Before and After the Ventilation Schedule (thesis Figures 5.28–5.30)

6.2 Shading and envelope

The design brief did not allow external shading on the façade, so three interior shading options were compared. Venetian blinds, with the highest reflectance, lower the cooling load from 78.01 to 61.82 kWh/m² while raising the heating load from 17.67 to 20.34 kWh/m², for a net EUI of 99.00 kWh/m² (−6.4%).

For the envelope, the thesis simulated four wall options, eight window U-values, six solar heat gain coefficients, and three roof options, using insulation thickness as a proxy for cost. It chose the point beyond which extra insulation brought little further saving (Table 3). Lowering the window SHGC mattered far more than lowering its U-value, and double glazing was rejected because it cut the total load by only 0.3% at roughly twice the cost.

Table 3. Selected Envelope Upgrades

Element	Baseline	Selected design
External wall	U 0.796	U 0.399 (45 mm phenolic foam)
Window	U 2.6, SHGC 0.75	U 1.4, SHGC 0.25 (low-e insulating glass)
Roof	U 0.493	U 0.395 (50 mm phenolic foam)

Ventilation, blinds, and the envelope together lower the heating load by 40.6% (to 10.50 kWh/m²), the cooling load by 36.1% (to 49.86 kWh/m²), and heating and cooling electricity by 36.9%, bringing the EUI to 88.10 kWh/m² (−16.7%).

7. Active Measures and Operation

Following GB 55015-2021, the heat pump COP was raised from 2.0 to 3.2, which alone cuts heating and cooling electricity by 37.5%. Lighting power density was lowered from 9 to 6 W/m² with efficient LEDs (−30.8% lighting electricity) and equipment power density from 15 to 9 W/m² with efficient office equipment (−40.0% equipment electricity). Lower internal gains raise the heating load (17.67 to 22.29 kWh/m²) and lower the cooling load (78.01 to 69.61 kWh/m²). Together, the active measures lower the EUI to 65.40 kWh/m² (−38.2%).

Occupant expectations matter as much as equipment. Tightening the comfort range from 18–28 °C (Class II) to 22–26 °C (Class I) would raise heating and cooling electricity from 47.84 to 83.24 kWh/m² and the EUI from 105.76 to 141.16 kWh/m²; in other words, accepting Class II comfort saves 42.5% of heating and cooling electricity and 25.1% of total energy. The thesis also proposes operational measures it could not simulate in DeST-C: a longer pre-cooling period before office hours, thermal storage to shift heat pump operation to off-peak hours, and sensor-based control through a building automation system.

8. Load Flexibility

To frame demand flexibility, the thesis classifies the loads of the envelope, lighting, equipment, and HVAC systems into three groups. Rigid loads, such as minimum lighting, minimum fresh air, emergency

systems, and Class II comfort, cannot be cut or moved. Reducible loads, such as solar gains, over-lighting, idle equipment, and over-conditioning, can be cut with better technology and control. Transferable loads, such as heat pump operation with storage, pre-cooling and pre-heating, high-power equipment tasks, and battery charging, can be moved to off-peak or solar hours. Because DeST-C cannot simulate these strategies, the thesis presents them as design recommendations rather than quantified results.

9. Integrated Design Results

Combining all passive and active measures lowers the EUI from 105.76 to 55.28 kWh/m² (−47.7%; Table 4). Heating and cooling electricity falls by 61.1% (47.84 to 18.60 kWh/m²), lighting by 30.8%, and equipment by 40.0%. Annual electricity use drops from 570,034 to 300,696 kWh, a saving of 269,338 kWh (Table 4).

Table 4. Energy Use Intensity by Design Stage (kWh/m² per year)

Design	Lighting	Equipment	Heating and cooling	Total	Change
Baseline	20.92	37.00	47.84	105.76	—
Passive measures	20.92	37.00	30.18	88.10	−16.7%
Active measures	14.48	22.20	28.72	65.40	−38.2%
Integrated design	14.48	22.20	18.60	55.28	−47.7%

Comfort improves as well. Without air conditioning, the open-plan office is now within 18–28 °C for 6,054 hours a year instead of 3,662 (+65.3%), and the hours above 30 °C fall from 3,942 to 695 (Figure 2). In winter the free-running temperature is somewhat lower than in the baseline, but the annual heating load still falls slightly, from 17.67 to 15.89 kWh/m².

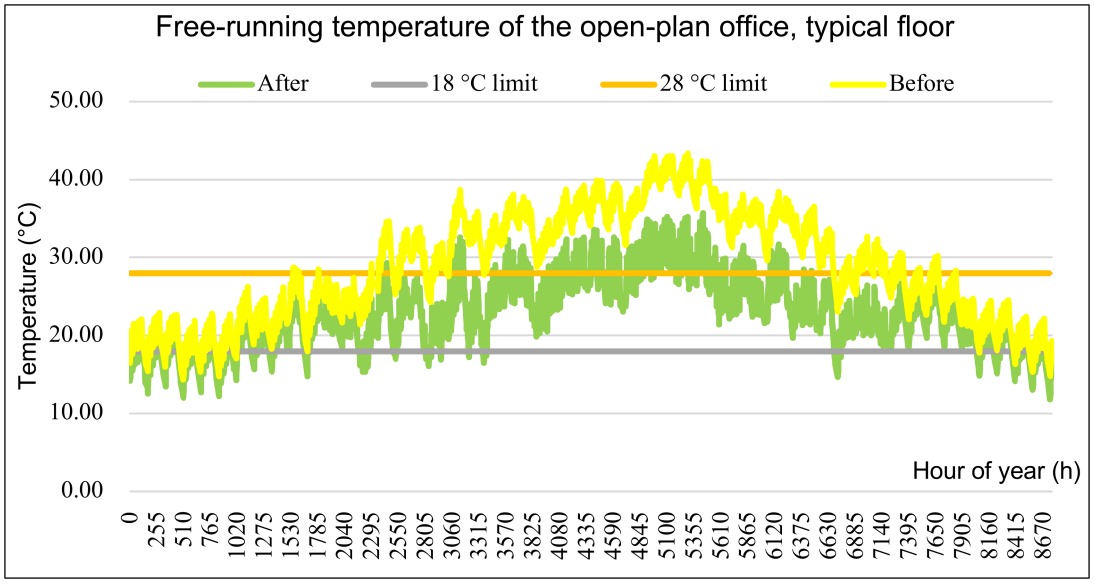


Figure 2. Hourly Free-running Temperature of the Open-plan Office on a Typical Floor over the Year, Before and After Optimization (thesis Figure 6.3)

On the heating design day, daily electricity falls by 41.8% and the morning peak by 46.7%. On the cooling design day it falls by 55.6%, the afternoon peak drops from 478 to 233 kWh (−51.2%), and almost no cooling is needed in the first two hours of the morning (Figure 3).

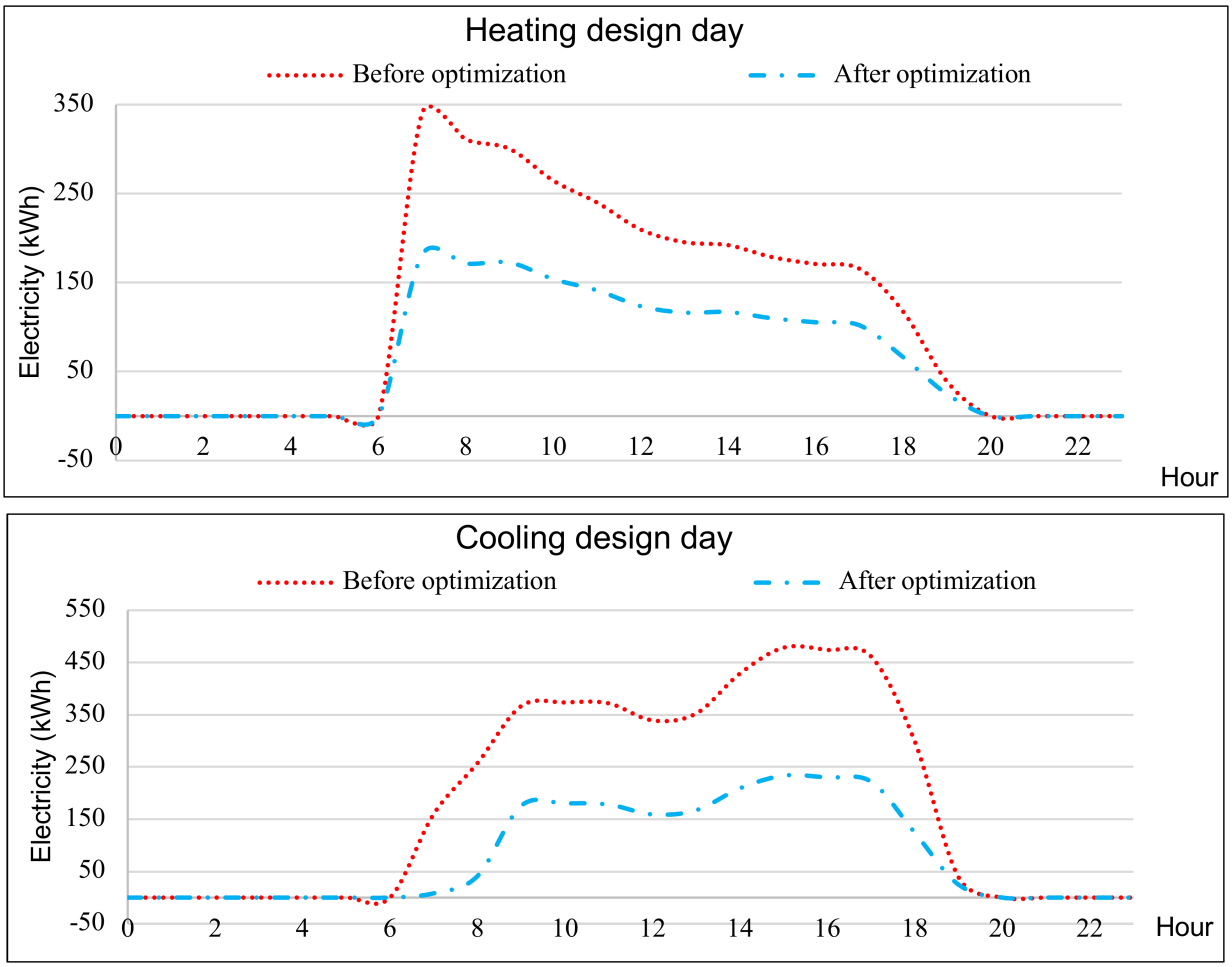


Figure 3. Hourly Building Electricity on the Heating (top) and Cooling (bottom) Design Days, Before and After Optimization (thesis Figures 6.1 and 6.2)

The rooftop system's 150,270 kWh covers all of the remaining lighting demand (87,094 kWh), and its surplus of 63,176 kWh supplies part of the equipment demand. Grid purchases fall to 150,427 kWh a year, and the annual electricity bill drops from about 467,400 CNY to 123,400 CNY, a saving of about 344,100 CNY.

10. Operational Carbon

Operational carbon savings were calculated with the method of GB/T 51366 and the average emission factor of the Southwest China grid:

$$\Delta C = \Delta E \times EF_{grid}, \qquad EF_{grid} = 0.5959\ kgCO_2/kWh$$

The 269,338 kWh saved each year avoid 160.5 t CO_2 (Table 5), or 30.08 kg CO_2/m^2 under the thesis's area convention. Heating and cooling contribute the largest share. The rooftop generation is not included in this figure; counted at the same factor, it would avoid a further 89.5 t CO_2 a year.

Table 5. Annual Electricity and Carbon Savings by End Use

End use	Electricity saved (kWh)	CO_2 avoided (kg)	Share
Lighting	38,743	23,087	12.8%
Equipment	77,485	46,174	29.3%
Heating and cooling	153,110	91,238	57.9%
Total	269,338	160,499	100%

As an illustration of scale, applying the same saving per square metre (44.8 kWh/m^2) to the office floor area Chongqing completed in 2024 gives about 111 million kWh and 66,000 t CO_2 a year; applying it to the floor area under construction gives about 2.11 billion kWh and 1.26 million t CO_2. These figures assume every building achieves the case-study savings and should be read as an upper bound.

11. Limitations

- **Simulation only.** All results come from simulation of a single building and are not validated against measured data.
- **Tools and weather data.** Ecotect and DeST-C are no longer updated, and the Ecotect weather data end in 2005, before the recent warming of Chongqing's summers. DeST-C also cannot represent several measures in detail, such as pre-cooling, storage, and smart control.
- **Simplified economics.** The photovoltaic estimate uses one annual irradiation value and a fixed performance factor, prices come from supplier quotes, and envelope choices use insulation thickness as a proxy for cost rather than a full life-cycle cost analysis.
- **Area convention.** The EUI sums end-use intensities over different floor areas, so its absolute values are not directly comparable with EUIs computed over gross floor area; the relative savings are.
- **Operational carbon only.** Embodied carbon in the added insulation, glazing, and photovoltaic modules is not counted.

12. Conclusion

For a typical office building in Chongqing's hot-summer, cold-winter climate, a combination of seasonal ventilation, interior shading, a moderately improved envelope, efficient heat pumps, lighting, and equipment, and a modest comfort standard can roughly halve operational energy use and raise free-running comfort hours by about two-thirds. Ventilation is a low-cost lever that works by matching air change rates to the season and to occupancy. Cooling dominates the load, so solar control matters more than lower window U-values. Even with Chongqing's weak solar resource, a horizontal rooftop array can supply all lighting electricity and pay back within ten years. Future work should validate the models with

measured data, update the weather files, and quantify the flexibility strategies, for example with storage and pre-cooling in a co-simulation framework.

Data and Code

The exported simulation results, the figures, and the full thesis (in Chinese) are available at https://github.com/JiaweiQi3/chongqing-office-low-carbon-design.

References

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[2] Ministry of Housing and Urban-Rural Development of China. GB 55015-2021, General Code for Energy Efficiency and Renewable Energy Application in Buildings.

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[5] Ministry of Housing and Urban-Rural Development of China. GB/T 51366-2019, Standard for Building Carbon Emission Calculation.

[6] ASHRAE. Standard 55, Thermal Environmental Conditions for Human Occupancy.